

Ahmed FAKHFAKH

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Data Scientist | Data Engineer | ML Engineer | Edge AI

Seeking a Permanent Position – Availability: October 2026

Data and AI Engineer from Polytech Nantes specializing in Edge AI and Generative AI: design of data pipelines on GCP, MLOps integration, multi-agent LLM and RAG systems, and optimization of Deep Learning models on NVIDIA Jetson via INT8 and TensorRT. Certified Google Cloud Professional Data Engineer.

CERTIFICATIONS

Google Cloud Certified – Professional Data Engineer | Associate Cloud Engineer | Cloud Digital Leader

SKILLS

- **Languages & Cloud:** Python, SQL, Bash, R, GCP BigQuery, Vertex AI, Cloud Run
- **AI, LLM & Computer Vision:** PyTorch, Machine Learning, Scikit-Learn, XGBoost, LangGraph, LlamaIndex, Qdrant, HuggingFace LoRA
- **Data Engineering & MLOps:** Apache Airflow, dbt, Spark, Kafka, MLflow, DVC, LangSmith, RAGAS
- **DevOps & Tools:** Terraform, Docker, CI/CD GitLab and GitHub Actions, FastAPI, Git, Linux

WORK EXPERIENCE

Edge AI Engineer Apprentice | OPmobility | Levallois-Perret, France **Oct. 2023 – Sep. 2026**

- **Edge AI Deployment & Profiling:** INT8 quantization and TensorRT compilation on NVIDIA Jetson. Reduced GPU inference latency from 865 ms to 188 ms, meaning a 4.6x acceleration reaching 5.3 FPS for a 10 FPS target frequency, and decreased RAM allocation from 10.2 GB to 3.5 GB.
- **R&D Deep Learning & Architecture:** Compressed AI architecture via Sparse networks and distillation. Reduced model weight from 53 MB to 17 MB while maintaining spatial detection probability Pd at 0.58.
- **Data Engineering & Software Engineering:** Implemented FIFO pipelines using Multiprocessing to cancel computational redundancy and integrated Kalman filters to smooth 3D trajectories.
- **Technical Stack:** Python, PyTorch, TensorRT, NVIDIA Jetson, Edge AI, Computer Vision, Multiprocessing.

SCIENTIFIC PUBLICATIONS

Real-Time 4D Radar Detection on Embedded GPUs through Compression and Quantization-Aware Training
- 2026 *Paper in preparation – Co-author* | Proof of hardware bottlenecks and evaluation of inference latency on embedded architecture.

Sparse-Candidate Attention for Radar Point-Cloud Generation on Embedded GPUs - 2026

Paper in preparation – Co-author | Design of a sparse attention architecture for radar point-cloud generation.

ADVANCED TECHNICAL PROJECTS

Multimodal Financial Agentic AI - Multi-Agent Architecture & MLOps **GitHub Link | 2025 – 2026**

- Designed a multi-agent system LangGraph orchestrating Big Data queries under BigQuery with self-correction, document RAG under ChromaDB and real-time APIs with an average latency of 1.2 s.
- **Security & Monitoring:** Implemented an AI firewall of type Guardrails anti-prompt injection and production cost observability via LangSmith.
- **Serverless Deployment & IaC:** Containerization under Docker on Google Cloud Run with asynchronous interface via Streamlit and infrastructure management via Terraform.
- **Stack & Tools:** GCP Cloud Run, BigQuery, Vertex AI Gemini, LLM, LangChain, ChromaDB, Terraform.

Modern Data Stack & Advanced RAG - RAGAS Evaluation

GitHub Link | 2025 – 2026

- Designed a complete Data Engineering pipeline of type ELT orchestrated by Apache Airflow, including data cleaning and transformation via dbt.
- Developed a production-ready Retrieval-Augmented Generation RAG system with LlamaIndex and vector database integration Qdrant hosting 10 000 indexed documents.
- Rigorous evaluation of RAG performance with a RAGAS Score of 92 % and hallucination detection using the RAGAS framework.
- **Stack & Tools:** Apache Airflow, dbt, LlamaIndex, Qdrant, RAGAS, Python.

EDUCATION

Data & AI Engineering Degree | Polytech Nantes, France

Oct. 2023 - Sep. 2026

Courses: Data Science, Machine Learning, Deep Learning, Computer Vision, MLOps.

Two-year Technical Degree in Computer Science | IUT de Bayonne et du Pays Basque, France **Sep. 2021 - Jun. 2023**

Courses: Mathematical Analysis, Algorithms, Object-Oriented Programming, SQL.

LANGUAGES

English TOEIC 830 | French Fluent | Arabic Native